SOEN 342: SOFTWARE REQUIREMENTS & DEPLOYMENT

Department of Computer Science and Software Engineering (CSSE)

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Project - Iteration I
System Modelling and Requirements Specification

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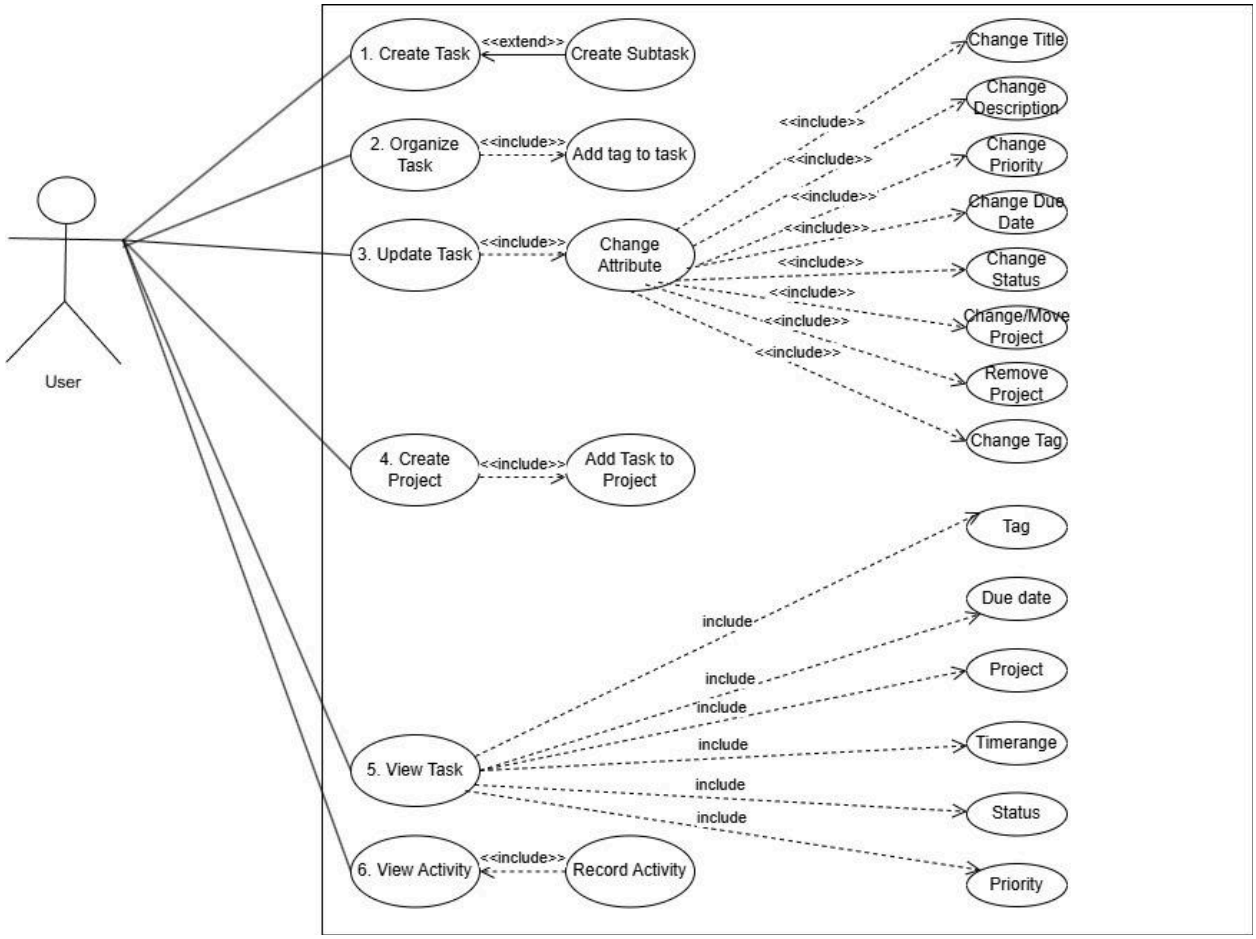
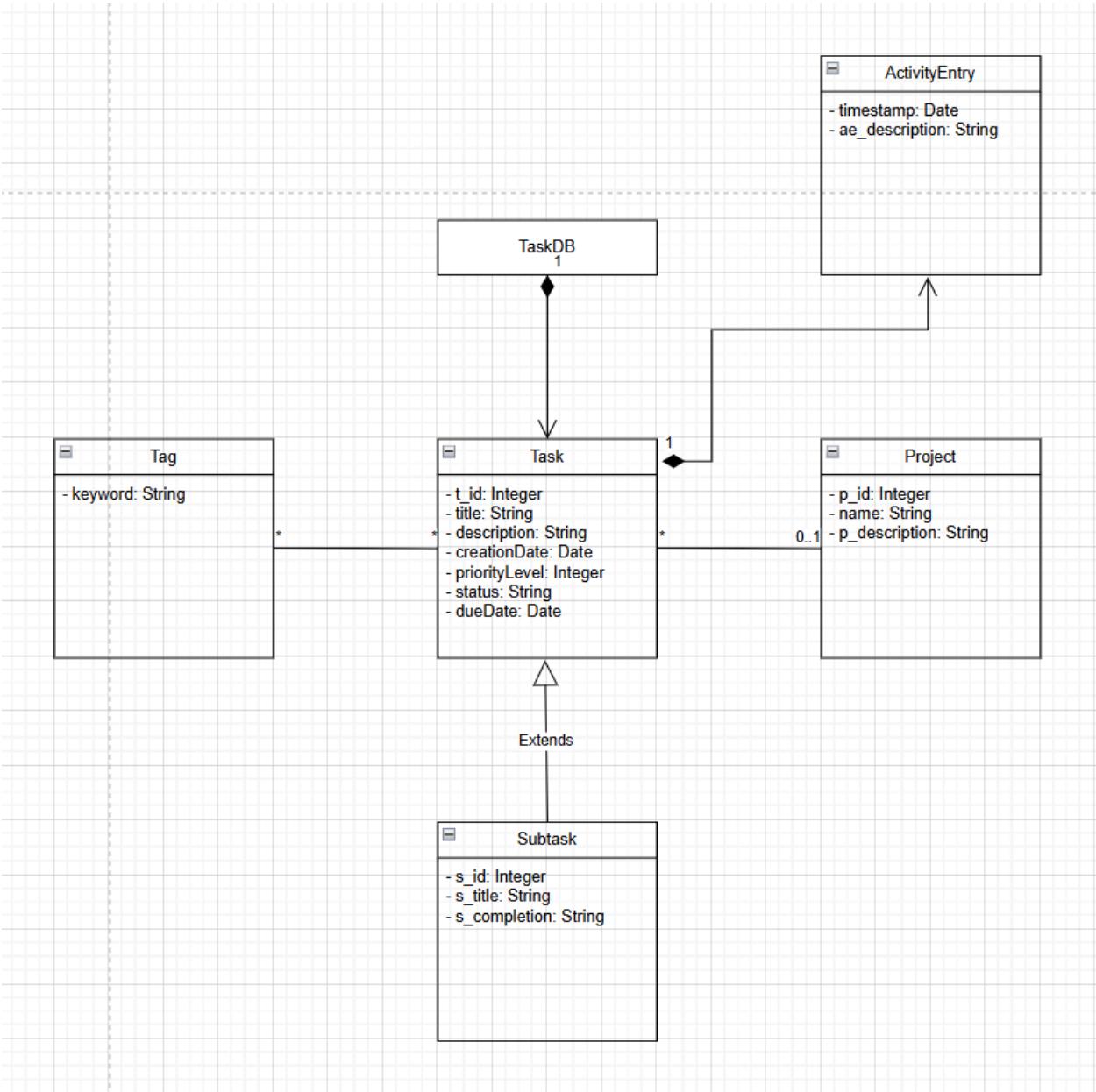
Iteration I

1. Use Case Diagram

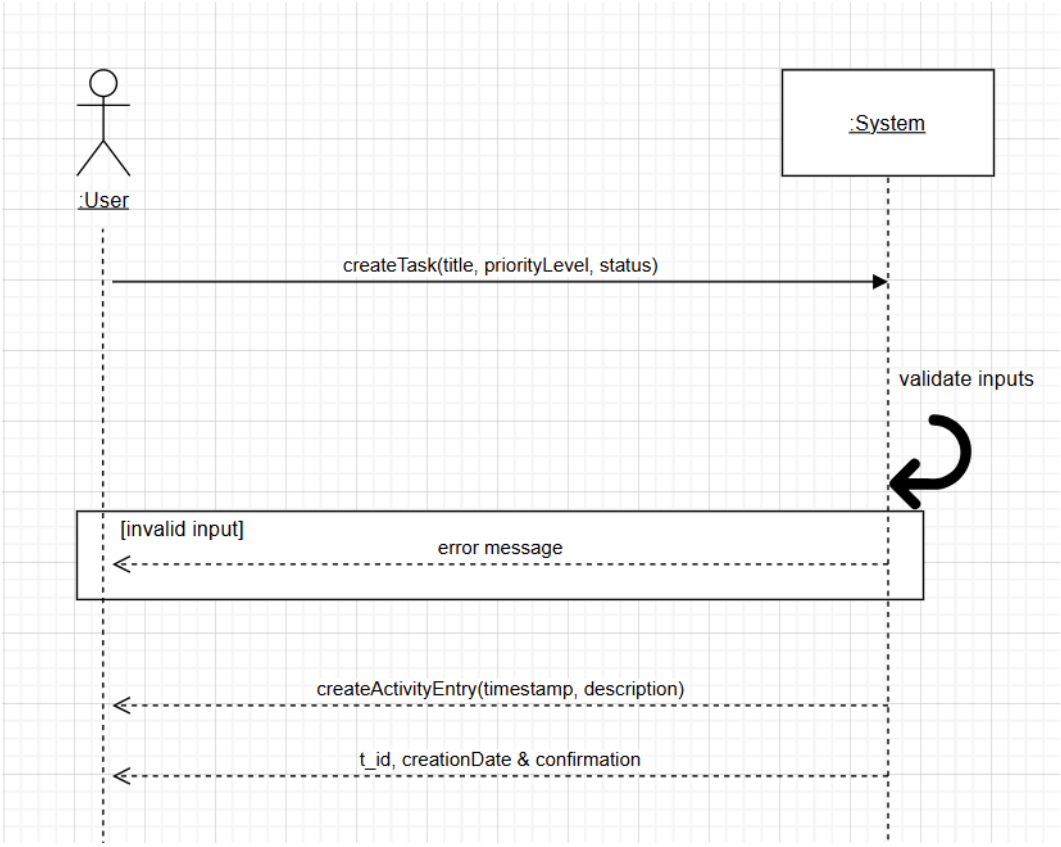


2. UML Domain Model

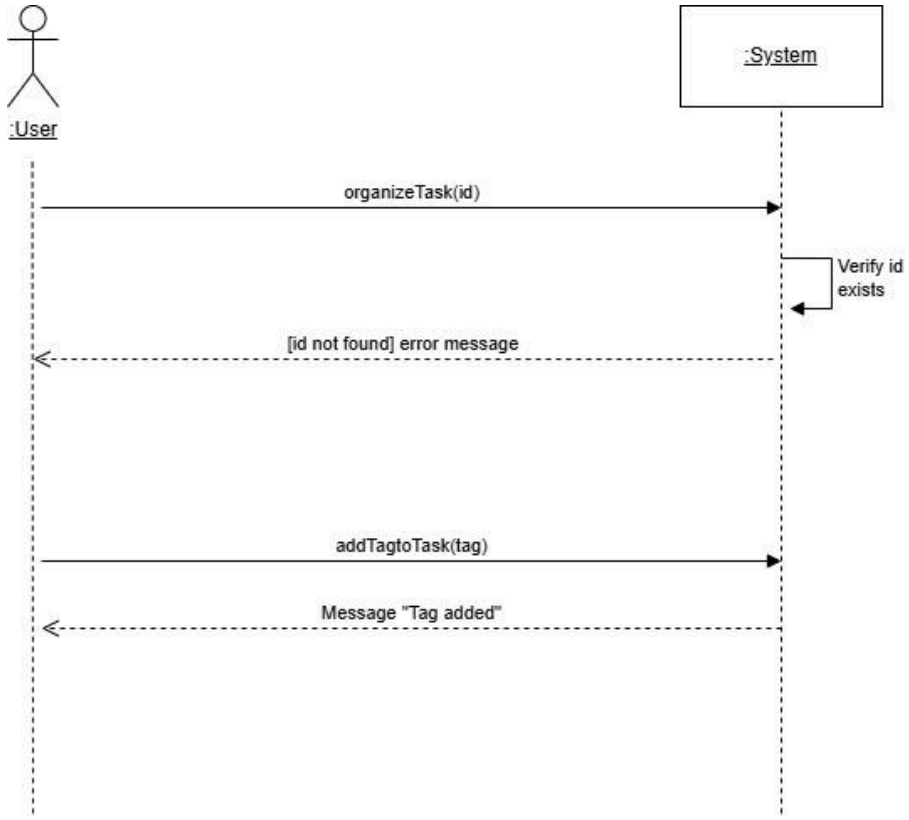


3. System Sequence Diagrams for Critical Use Cases

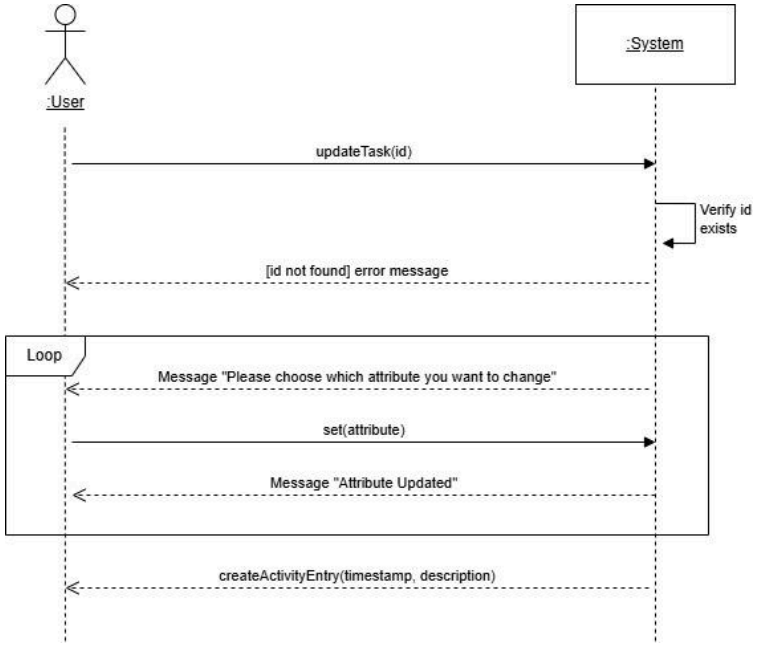
3.1 Create Task



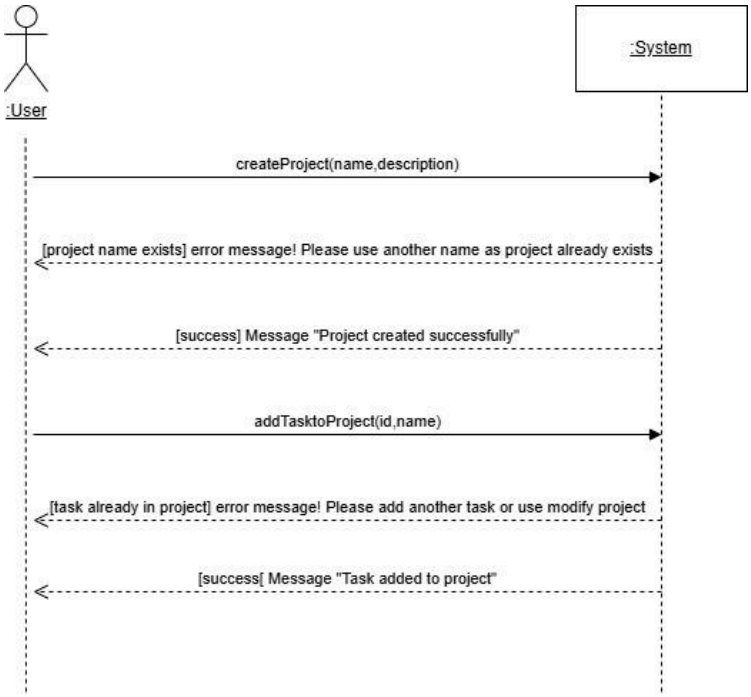
3.2 Organize Task



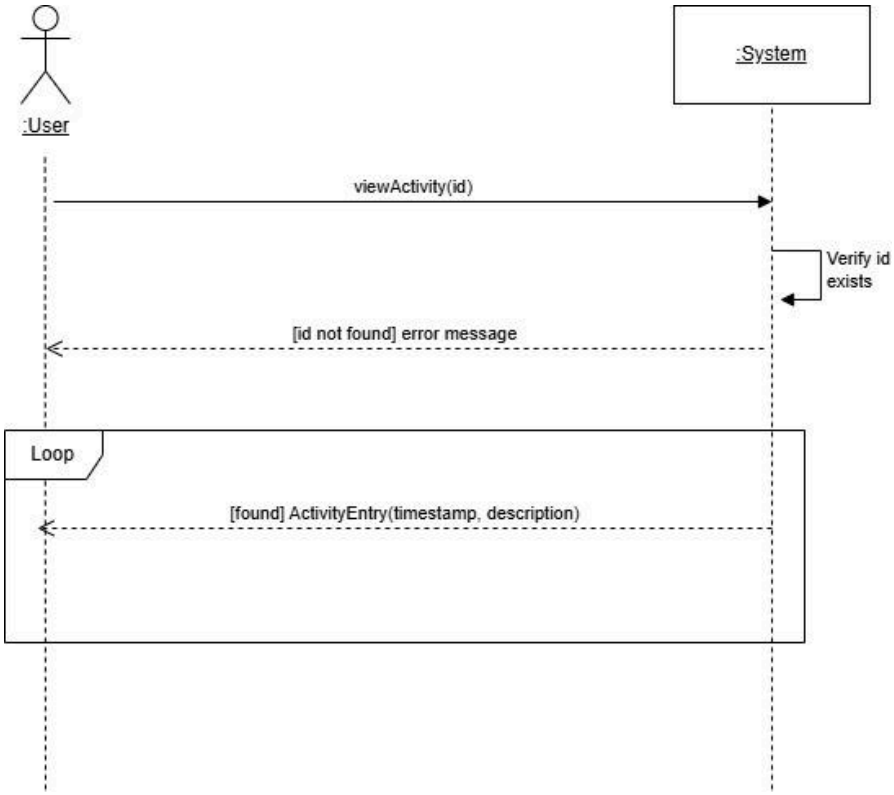
3.3 Update Task



3.4 Create Project



3.5 View Activity



4. Fully-Dressed Scenario for Use Case Create Task

Use case UC1: Create Task

Primary Actor: User

Stakeholders and Interests: User - Wants to save tasks efficiently and simply

Preconditions:

- User is logged in

- Task inputs are valid

Postconditions:

- Task is created (with id, creation date)

- Activity entry is created (with timestamp, description)

Main Success Scenario:

The user selects the Create Task option from the menu.

The system asks the user to enter the title of the task.

The user types in the task title and presses Enter.

The system validates the input.

Repeat step 2 to step 4 for the priority level and the status of the task.

The system creates an activity entry with a timestamp and a description of the action executed.

The system sends a confirmation message to the user with the task id and creation date.

Extensions:

If invalid input entered, indicate error.

5. System Operations & Operation Contracts

5.1 Create Task

Use case: Create Task

System Operations: createTask(title, priorityLevel, status)

Contract CO1: createTask

Operation: createTask(title: String, priorityLevel: Integer, status: String)

Cross References: Use case Create Task

Preconditions: The title, the priority level, and the status are valid user inputs.

Postconditions: A task instance t was created, attributes of t (title, priority and status) have been set with values by data from the user, t is added to TaskDB, an activityEntry is created regarding the task.

5.2 Organize Task

Use case: Organize Task

System Operations: organizeTask(id)

Contract CO2: organizeTask

Operation: organizeTask(id: Integer)

Cross Reference: Use case Organize Task

Preconditions: Task with TaskId exists, tagName is valid

Postconditions: If a tagName exists it is used if not it is created, selected task is associated with the tag, task now includes the tag in its collection, confirmation message is returned

5.3 Update Task

Use case: Update Task

System Operations: updateTask(id)

Contract CO3: updateTask

Operation: updateTask(id: Integer)

Cross Reference: Use case Update Task

Preconditions: Task with TaskId exists, the specified attribute is a valid modifiable attribute

Postconditions: The specified attribute is updated, an activityEntry is created regarding the task, confirmation message is returned

5.4 Create Project

Use case: Create Project

System Operations: createProject(name, description)

Contract CO4: createProject

Operation: createProject(name: String, description: String)

Cross Reference: Use case Create Project

Preconditions: the project name is a valid String, the project name does not match any existing project names

Postconditions: a new project instance is created, project name is set to the provided name. Project description is set to the provided description, Specific Task is associated with the project, a confirmation message is returned

5.5 View Activity

Use case: View Activity

System operations: viewActivity(id)

Contract CO5 viewActivity

Operation: viewActivity(id: Integer)

Cross Reference: Use case View Activity

Preconditions: task with taskId exists

Postconditions: the task with taskId is returned, all activityEntry associated with task is returned to user